

Everything you need to know about the Sennheiser Spectera, world's first bidirectional digital wireless ecosystem

In conversation with **Vipin Pungalia**, Country Manager & Director – Sales Pro Audio at Sennheiser India

Dhriti Datta | dhriti@digit.in

In the fast-evolving landscape of professional audio, Sennheiser continues to attempt to set groundbreaking standards, this time with the launch of Spectera — the world's first bidirectional digital wireless ecosystem for microphones and monitoring. Designed to address audio engineers' complex challenges in live performance and studio environments, the Sennheiser Spectera combines new-age technology with practical solutions to simplify audio setups, enhance transmission reliability, and improve workflow efficiency. In our interview with Vipin Pungalia, Country Manager & Director – Sales Pro Audio at Sennheiser India, we explore how Spectera's innovative use of wideband RF channels and bidirectional communication is poised to transform large-scale productions. From reducing interference to accommodating extensive multi-channel setups in a compact unit, Spectera represents a new era for



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Country Manager & Director – Sales
Pro Audio at Sennheiser India

wireless audio, rooted in Sennheiser's rich legacy of innovation. In this insightful interview, Vipin Pungalia explains everything you need to know about the Sennheiser Spectera.

Q Sennheiser has introduced the Sennheiser Spectera, the world's first bidirectional digital wireless ecosystem for microphones and monitoring. Could you elaborate on how this innovation addresses the current challenges faced by audio engineers in live performance and studio environments and how it improves on equipment before this?

A The Spectera ecosystem addresses key challenges faced by audio engi-

neers in multichannel live audio and (broadcast) studio environments by simplifying audio set-ups considerably, giving the operator full control of the audio system, greatly improving transmission reliability, and reducing the footprint of wireless audio systems massively.

Where with traditional narrowband systems, IEMs and microphones need to be operated in separate TV channels with a guard band in between, Sennheiser Spectera has bidirectional audio links and bidirectional control data in the same RF channel. Operators have full control and can select latency, channel count and range via various Audio Link Modes.

Furthermore, Spectera can accommodate up to 32 inputs and 32 outputs in a single 1U rack-mount device. The time and space you save during set-up and teardown, and in the warehouse and during transit, are enormous. Where you currently need two bodypacks with narrowband systems, Spectera will just require one.

Additionally, RF fading and interference, common issues that can cause audio dropouts, are mitigated by Spectera's use of a wideband RF channel, enhancing diversity up to 40-fold for an 8 MHz channel. This minimizes the risk of signal loss, ensuring more reliable performance, which is crucial in any live audio environment like concerts and broadcasts.

Mind you, there will always be a need for narrowband systems, too, for example with smaller audio set-ups.